

Siddarath Vats

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Education

Southern Methodist University, Dallas, Texas - Master of Science, Computer Science-AI
Adani University, Gujarat, India - Bachelor of Engineering, Electrical Engineering

May 2025
Jun 2021

Technical Skills

Programming languages: Python; SQL; R

Database Environments: MySQL workbench; Google Big-query editor; GCP cloud SQL; Redshift

Frameworks: LangGraph,; Agent Development Kit; LangChain ;Hugging Face; Scikit-learn; Pytorch; TensorFlow; keras; PySpark(beginner); Selenium

Cloud Platforms & Services: **GCP:** Big Query; Vertex AI; Cloud Scheduler, **AWS:** SageMaker; Bedrock; Redshift; Athena

Supporting Applications: GitHub; Virtual Machines; Linux- GPU clusters; Docker

Professional Experience

Research Assistant, SMU, Lyle School of Engineering, Dallas, Texas

July 2024 – Present

- Built a multi-agent architecture using LangGraph to extract Social Determinants of Health (SDOH) from the MINIC-3 clinical notes dataset, orchestrating large language models such as DeepSeek, Mistral-7B, and LLaMA 3.3; the resulting annotations were later used to improve downstream medical deep learning models like BioClinicalBERT, Med-BERT across multiple patient cohorts.
- Deployed models locally via **Ollama** and optimized model inference for speed, leveraging **NVIDIA A100 GPUs** on the **SMU SuperPOD HPC cluster** for efficient, large-scale inference. Stored structured SDOH annotations in **TimescaleDB**, enabling **temporal analysis**, cohort tracking, and visualization of social trends in patient health data.

Junior Data Scientist, Adani Green Energy Ltd, Ahmedabad, India.

Jun 2021 – Aug 2023

- Designed and deployed advanced AI/ML solutions using deep learning algorithms in custom pipelines on Google Cloud with Vertex AI for solar plant performance monitoring, improving operational performance ratio (PR) by **~2%** by following the procedure of MLOps.
- Engineered ETL pipelines to fetch data from BigQuery, Redshift, S3, and Google Cloud Storage, ensuring smooth data flow for model training and analysis via automated deployment pipelines under CI/CD norms.
- Developed end-to-end automated systems for data ingestion, transformation, and analysis using Python, SQL, and Power BI, enabling real-time DSM dashboard updates.
- Reduced penalty costs **by ~10%** through predictive analytics and algorithmic optimization, while leading the development of interactive Power BI dashboards for Solar and Wind Portfolios, enhancing insights for **~5,000 employees**.
- Automated and scaled analytics pipelines to support predictive maintenance and data-driven decision-making across a **~9 GW renewable energy portfolio**.
- Honored with a **silver coin and spot recognition award** for dashboard development and enabling full-stack teams with accurate data integration.

Research Work

- Automated financial report analysis using NLP and LLMs to extract data and identify trends from SEC EDGAR and Senate disclosures, developing predictive models for financial compliance risks for a potential NLP conference publication.
- Contributed to a project evaluating **low-SES students** using Reddit data; used **ETL pipelines on Google Cloud (Cloud Storage, BigQuery, and Cloud Functions)** and deployed **GPT-3.0** for narrative extraction and insight generation.
- Analyzed model performance using confusion matrices and visualized results with **Google Data Studio** and **PowerBI**; explored integration of ML-based fault prediction with robotic intervention for real-time power restoration.
- Reviewed existing technologies for differentiating between three-phase faults and power swings; published findings in *Nova Science Publishers*.[Link to publication](#)

Activities

- Member of SMU Statistics and Data Science Club – participated in data challenges, workshops, and collaborative projects.
- Member of SMU Chess Club – engaged in tournaments and strategy sessions.
- Currently building an AI-powered chess app with the SMU Chess Club that helps players practice chess coordinates through interactive drills and real-time feedback.
- Initiated the design of an AI-driven tax compilation for a firm system utilizing LangGraph and OpenAI's ChatGPT API. The project focuses on creating modular agents for tasks such as data ingestion, validation, and report generation, aiming to streamline financial reporting processes.

Certifications

- AWS Certified Machine Learning – Specialty(In progress).
- Databricks Certified Associate Developer for PySpark (In progress)
- UT Austin- Diploma in AI powered by Great Learning
- Fundamentals of Machine Learning- Andrew Ng